

Predicting the efficacy of malaria vaccines across the life-course

Supplementary Notebook 2. Averted deaths in a cohort vs. a population

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Previous vaccine modelling studies have modelled intervention campaigns in stable, multi-cohort populations over a fixed time interval, usually 15 years, with constant and equal birth and death rates (e.g. Penny et al. (2016); Schmit et al. (2024)). Two reasons have been given to justify a fixed endpoint of 15 years: to be comparable to the timeframe of the Global Technical Strategy for Malaria (“World Health Organization. Global Technical Strategy for Malaria 2016–2030, 2015.” 2015; Penny et al. 2016), to capture the negative effects of infections shifting to older children - meaning rebound in cases and the higher risk of severe malaria-associated mortality in older children (Reyburn et al. 2005; Penny et al. 2016; Schmit et al. 2024).

Unfortunately, no data exists to refute or support the assumption that the detrimental effects of rebound will be negligible after 15 years due to the practical difficulties of obtaining such data (Pemberton-Ross et al. 2015). Even assuming that rebound is negligible after 15 years, it is only the children at the start of the campaign that will have experienced the full detriment of rebound; younger children will have experienced only a fraction of its negative effects. Thus, calculating efficacy at some fixed endpoint will truncate and reduce the magnitude of the detrimental effects of rebound at the population level. In addition, the youngest children are over-represented in the calculation of vaccine efficacy. For example, after 15 years there are 15 times more newborns in total than 15-year-olds because there have been 15 years of birth, but only one cohort has aged to 15 years old. And it is the youngest children that benefit most from vaccination due to its waning efficacy. For these reasons, modelling a vaccine campaign in a population over a fixed time period over-estimates averted cases and vaccine efficacy.

Vaccine trials, on the other hand, are longitudinal studies in single, multi-age cohorts and efficacy is calculated for this single cohort relative to an unvaccinated control cohort. This definition directly expresses a vaccine’s efficacy as the percentage reduction in risk of morbidity and mortality over the cohort’s lifetime.

Figure 1 compares averted deaths over either in a multi-cohort population (orange bars) or a single cohort (blue bars) for R21 given to 100,000 5 month olds. Note the different meaning on the x -axis. The single cohort completes three rounds of vaccination then receives no more doses; time on the x -axis represents time since the third and last vaccine dose. For the multi-cohort population, each year a new cohort of 5 month olds complete three rounds of vaccination; time on the x -axis represents time since the start of the vaccine campaign. Hence vaccine-induced protection in the population is being “topped-up” every year, whereas in the single cohort, protection gradually decays from a maximum at $t = 0$.

Averted deaths are positive for the first five years but decline due to the waning efficacy of R21. After five years rebound occurs (negative averted deaths) but is less for the population. After 20 years cumulative averted deaths has plateaued. Vaccination in the population averted 13 deaths, whereas vaccination in the cohort caused 56 deaths.

This figure also demonstrates that imposing a 15 year endpoint would fail to count deaths after 15 years which would under-estimate deaths and over-estimate efficacy.

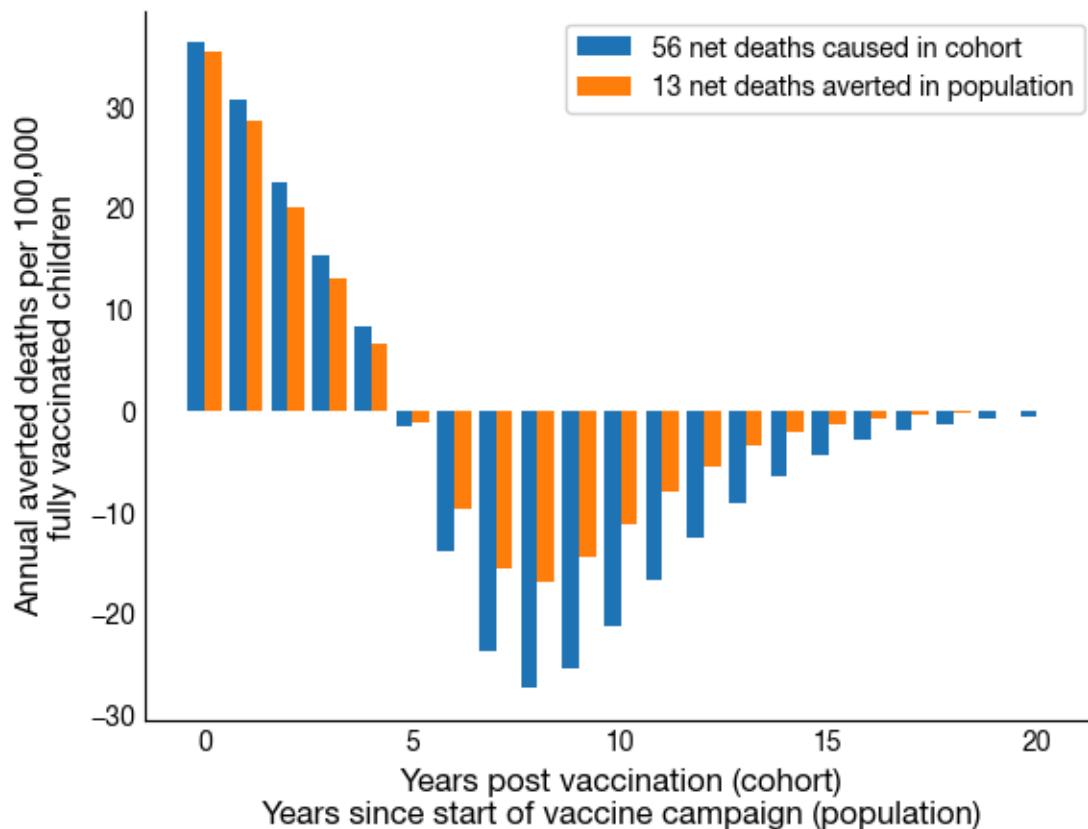


Figure 1: Annual averted deaths per 100,000 R21-fully vaccinated five month old children, comparing a multi-cohort population (orange) and a single cohort (blue) and assuming 1 blood infection every 2 years. The multi-cohort population assumes a vaccine campaign that vaccinates 100% of five month old children each year for 20 years, while the single cohort assumes a one-time vaccination of 100% of five month old children at the start of the study. The total number of deaths averted over the study period is shown in the legend.

Bibliography

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